BRENT WILLIS

SENIOR FULL STACK & AI ENGINEER

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SUMMARY

Senior Full Stack & AI Engineer with 10+ years of experience building and shipping production software across AI, healthcare, e-commerce, and Web3. Strong backend foundation in Python, Node.js, APIs, databases, and cloud-based systems, with recent focus on production AI agents, chatbots, and multi-step workflows. Experienced in integrating AI capabilities with real business systems, designing reliable backend services, and taking products from requirements and architecture through deployment, debugging, and production support.

TECHNICAL SKILLS

AI & Automation: AI Product Integration, AI Agents, Chatbots, Workflow Automation, Data Processing, Classification Models, Model-Driven Applications

Languages: Python, TypeScript, JavaScript, SQL, PHP

Backend & APIs: Node.js, Express.js, NestJS, Django, Laravel, REST APIs, GraphQL, WebSockets, Microservices, API Integration

Databases: PostgreSQL, MySQL, MongoDB, Relational Data Modeling, Query Optimization

Cloud & DevOps: AWS, Google Cloud Platform, Docker, Kubernetes, Cloud Foundry, Nginx, Git, GitHub, CI/CD

Frontend: React, Next.js, Redux, Context API, React Native, Tailwind CSS, Material UI

Web3: MetaMask, WalletConnect, Blockchain/Web3 Integrations

Engineering: System Design, Backend Architecture, Authentication, Testing, Debugging, Performance Optimization, Technical Leadership

PROFESSIONAL EXPERIENCE

Senior AI Engineer | GenAI.Labs | United States / Remote | Mar 2023 – Present

- Build production AI agents, custom chatbots, and AI-powered web applications using Python, Node.js, React, Next.js, PostgreSQL, and LLM APIs, helping move AI capabilities into real business applications.
- Design multi-step AI workflows that process user requests, retrieve business data, call external APIs, and automate repetitive client operations.
- Develop LLM-powered chatbots for customer support and internal knowledge access, helping teams respond faster and handle more requests efficiently.
- Design backend orchestration with Python and Node.js, including tool calling, background jobs, retries, validation, and failure recovery to support reliable AI workflows.
- Integrate AI capabilities with existing business systems through REST APIs, GraphQL, webhooks, authentication, databases, and third-party services, enabling AI features to operate within existing workflows.
- Resolve production reliability issues such as malformed model outputs, API failures, duplicate jobs, and inconsistent responses through structured outputs, retry logic, fallback handling, and monitoring.
- Build React/Next.js interfaces for chat, workflow status, AI-generated content review, and operational dashboards, improving usability and visibility into AI-driven workflows.
- Own projects from client requirements and architecture through implementation, deployment, debugging, and production support, helping transition AI solutions from prototypes into maintainable products.

Senior Backend Engineer | Mount Carmel Health System | United States / Remote | Jan 2021 – Jan 2023

- Built backend application features supporting healthcare data workflows, with emphasis on APIs, data integration, reliability, and secure service-to-service communication.
- Developed RESTful services and data-processing components that improved interoperability between application interfaces and backend systems, supporting more reliable healthcare workflows.
- Implemented real-time data flows using WebSockets and integrated cloud APIs with customer-facing and operational interfaces, enabling timely movement of application data.
- Applied data validation, error handling, and integration patterns to support reliable healthcare platform workflows and reduce integration-related failures.
- Contributed to React frontend development while maintaining a primary focus on backend services, integration behavior, and production support.

Web3 Application Engineer | CryptoSlam Inc | United States / Remote | Dec 2018 – Dec 2020

- Built and improved data-heavy user interfaces using React, Next.js, TypeScript, and Material UI for blockchain and digital-asset workflows.
- Integrated MetaMask and WalletConnect for wallet connectivity and Web3 user flows, enabling users to interact with blockchain functionality through the application.
- Integrated REST APIs and asynchronous data flows into responsive, cross-browser application experiences, supporting reliable access to application data.
- Worked with Node.js services and API integration layers supporting authentication, routing, and client-side data requirements.
- Contributed across full-stack implementation while focusing primarily on frontend architecture, usability, performance, and blockchain integration, helping deliver responsive Web3 application experiences.

Senior Software Engineer | Amazon | United States / on-site | Feb 2016 – Feb 2018

- Delivered full-stack client applications across frontend and backend using React, Next.js, Node.js, REST APIs, and service integrations.
- Built responsive, data-driven interfaces connected to backend services through authentication, routing, request handling, and reusable API contracts, supporting consistent application behavior.
- Refactored backend payment and integration services to improve maintainability, scalability, and reliability.
- Created Node.js proxy and integration tooling to simulate external API responses and manage authentication, headers, and routing during development, supporting more efficient integration testing.
- Collaborated with client stakeholders and engineering teams on requirements, implementation, debugging, and production-ready releases, helping deliver stable application functionality.

Senior Software Engineer | Paymore | United States / on-site | Sep 2012 – Dec 2015

- Built and maintained e-commerce application functionality supporting product discovery, catalog data, search, customer workflows, and online purchasing.
- Developed backend services and REST APIs using PHP/Laravel with relational database integration and structured data processing, supporting core commerce workflows.
- Implemented frontend functionality using HTML, CSS, JavaScript, jQuery, AJAX, and JSON, connecting shopping interfaces with backend services.
- Cleaned and transformed unstructured product data into relational datasets used by search and commerce workflows, improving the usability and consistency of product information.
- Supported debugging, integration, performance, and maintenance across frontend and backend systems, helping keep e-commerce functionality reliable.

EDUCATION

Bachelor's Degree in Computer Science | The Ohio State University | Sep 2008 – Aug 2012

Major: Computer Science | Minor: Machine Learning